

Fractional-Order Adaptive Motion Magnification: Phase-Reliability Weighting for Noise-Constrained Video Amplification

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Abstract—Eulerian video amplification boosts sub-pixel motion by band-pass filtering per-pixel intensity traces and applying a uniform gain. That gain ignores local structure, so sensor noise is amplified together with the signal, especially in textureless regions where the monogenic phase is unreliable. We propose FrAM (Fractional-order Adaptive Motion Magnification), a pipeline developed first offline and then as a causal stream. It replaces the constant temporal gain with a Grünwald–Letnikov derivative of fractional order, giving continuous control over high-frequency emphasis, and replaces the uniform spatial gain with a per-pixel weight derived from the local amplitude of the monogenic signal. On a controlled synthetic sequence split into textured and flat halves, FrAM matches the amplification of the Eulerian baseline while keeping flat-region temporal noise at the input level. The reduction holds across an eightfold range of input noise levels. Real videos show improved spatial selectivity and lower background noise in every case. The causal reformulation cuts the per-frame cost by two orders of magnitude, reaching 69 fps at 640×480.

Index Terms—video magnification, fractional calculus, Grünwald–Letnikov derivative, Riesz transform, monogenic signal, phase-based processing

I. INTRODUCTION

A camera records many motions that are too small for a human observer to see: the pulse-induced undulation of skin, the vibration of a mechanical structure, or the sway of a loaded beam. The purpose of *video magnification* is to make these imperceptible displacements visible without attaching physical sensors to the scene.

The Eulerian formulation introduced by Wu *et al.* [1] recasts the problem in a way that avoids the difficulties of tracking. Instead of following points across frames—a Lagrangian strategy that requires sub-pixel correspondence and is fragile at fine scales—it monitors the intensity evolution at *fixed* pixel locations. For a small displacement $\delta(t)$, a first-order Taylor expansion gives

$$I(x + \delta(t), t) \approx I(x, t) + \delta(t) \nabla I(x, t), \quad (1)$$

so motion manifests as an intensity fluctuation proportional to the local spatial gradient. What was a spatial estimation problem becomes a temporal filtering problem: decompose each frame into spatial-frequency bands with a Laplacian pyramid,

temporally band-pass each band to isolate the frequency of interest, multiply by a gain α , and reconstruct.

The limitation of this approach is that (1) gives no indication of *where* the amplification should be applied. The gain is the same for every pixel, so the temporal band-pass amplifies sensor noise and genuine motion with identical strength. Because noise spans the whole spectrum, some of it always lies inside the pass-band. In regions that lack structure the output is therefore mostly amplified noise, producing the temporal mottling commonly reported in the literature. In practice it is this artefact, not any formal upper bound, that restricts the useful gain.

Phase-based methods [2], [3] improve the situation by amplifying the *phase* of a complex steerable or Riesz pyramid rather than its amplitude, shifting structure without changing contrast. Their noise behaviour is markedly better, but an important precondition is often left unstated: the local phase of the monogenic signal [7] is well defined only where the local amplitude is non-negligible. Where $A \rightarrow 0$ the phase collapses into noise, and amplifying it reintroduces the very artefact the phase formulation was designed to avoid.

This paper builds on that observation and makes two contributions:

- 1) A **fractional-order temporal filter** based on the Grünwald–Letnikov derivative, applied after band-pass filtering with the frequency response $|(1 - e^{-i\omega})^\nu|$. The order $\nu \in [0, 1]$ continuously interpolates between the identity ($\nu = 0$) and a first temporal difference ($\nu = 1$), setting how strongly high temporal frequencies—and therefore noise—are emphasised.
- 2) A **phase-reliability adaptive gain** that modulates the amplification per pixel according to the local monogenic amplitude. Where the phase is uninformative the gain is reduced. The cutoff σ is chosen as a percentile of the amplitude distribution of each band, so the method adapts to the content without manual tuning.

The evaluation uses a synthetic sequence whose ground-truth displacement is known, which makes it possible to measure the achieved amplification factor rather than relying solely on visual inspection. An ablation is included to disentangle the two contributions.

II. RELATED WORK

A. Amplitude-based magnification

Eulerian Video Magnification [1] established the pipeline just described. Its simplicity and low computational cost have made it the baseline method, and it remains effective for colour magnification, where the quantity of interest is an intensity change rather than a displacement. Its main weakness is the linear growth of noise with the gain.

B. Phase-based magnification

Wadhwa *et al.* [2] replaced the Laplacian pyramid with a complex steerable pyramid [8] and amplified local phase, achieving much better noise performance at higher computational cost. The Riesz pyramid [3] lowered that cost by approximating the quadrature pair with the Riesz transform, a two-dimensional extension of the Hilbert transform, and reached real-time speeds. The multipliers that define the Riesz transform belong to the family of singular integrals studied in fractional calculus [6], a link that motivated the present formulation.

C. Fractional calculus in image processing

Fractional-order operators have been used for image denoising, edge detection, and texture enhancement [4], [5]. In those applications the non-integer order provides a trade-off between edge sharpening and noise suppression that integer-order derivatives cannot offer. Almost all such work is *spatial*. Applying fractional differentiation along the *temporal* axis of a video-magnification pipeline, as we do here, does not appear to have been explored before.

III. METHOD

A. The Grünwald–Letnikov operator

Among the many equivalent definitions of a fractional derivative [6], the Grünwald–Letnikov form is the most convenient to discretise directly:

$$D^\nu f(t) \approx h^{-\nu} \sum_{k=0}^K w_k f(t - kh), \quad w_k = (-1)^k \binom{\nu}{k}, \quad (2)$$

with coefficients computed through the numerically stable recurrence

$$w_0 = 1, \quad w_k = w_{k-1} \frac{k-1-\nu}{k}. \quad (3)$$

Two properties of (2) are central to the method.

Continuous interpolation. For $\nu = 1$ the coefficients become $[1, -1, 0, \dots]$, which is the classical backward difference $x[n] - x[n-1]$; for $\nu = 2$ they become $[1, -2, 1]$, the classical second-order backward difference. The frequency-domain expression follows from the shift property of the discrete Fourier transform: a one-sample delay, $x[n-1]$, corresponds to multiplication by $e^{-i\omega}$, so the first-order backward difference has response

$$H_1(\omega) = 1 - e^{-i\omega}.$$

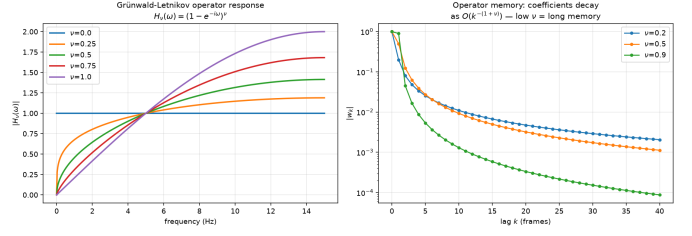


Fig. 1. Left: magnitude of the Grünwald–Letnikov response (4), showing the continuous interpolation between the identity ($\nu = 0$) and a first derivative ($\nu = 1$). Right: decay of the coefficients w_k as a function of memory lag k ; smaller ν implies a longer effective memory. Here k indexes past frames in the GL sum and measures the operator’s temporal reach, not a group-delay or control-loop reaction lag.

Raising this response to an arbitrary real power gives the continuous-order generalisation

$$H_\nu(\omega) = (1 - e^{-i\omega})^\nu \sim (i\omega)^\nu, \quad (4)$$

where the approximation uses $e^{-i\omega} \approx 1 - i\omega$ for small ω . Integer values recover the usual finite differences, while non-integer ν defines operators with no integer-order counterpart. Since $(i\omega)^\nu = |\omega|^\nu e^{i\nu\pi/2}$ for positive frequencies, the operator has a gain proportional to $|\omega|^\nu$ and a phase shift of $\nu\pi/2$. Thus the order gives continuous control over the emphasis of high-frequency components (Fig. 1, left).

Long memory. The Grünwald–Letnikov weights decay algebraically, $w_k = O(k^{-(1+\nu)})$, rather than exponentially [9]. The operator is therefore non-local in time, and the memory grows as ν decreases. Because the tail is algebraic there is no sharp cutoff; truncation is a matter of tolerance rather than convergence. The implementation estimates the needed length by choosing the smallest K for which the residual weight mass inside a 512-term window falls below a relative tolerance of 10^{-3} . This yields $K = 503$ at $\nu = 0.2$ and only $K = 124$ at $\nu = 0.9$ (Fig. 1, right). At $\nu = 0.2$ the residual only crosses the tolerance near the edge of the window, so no much shorter memory satisfies the criterion. This long memory is accepted by the offline formulation, which works with the full temporal history; Sec. VII and Table VIII show that a causal implementation needs far less.

B. Fractional band-pass filter

Because the fractional operator emphasises high frequencies, applying it directly to a broadband signal would also boost out-of-band noise. The temporal processing is therefore split into two stages. A second-order Butterworth band-pass, implemented as a biquad (two poles and two zeros; Appendix A explains why the order-1 design used here produces such a section), restricts each band-level time series to the interval $[f_{lo}, f_{hi}]$. It is applied with zero-phase forward-backward filtering so that no group delay is introduced. The fractional derivative (2) is then applied to the band-limited signal. With the sampling interval normalised to one frame, the $h^{-\nu}$ factor in (2) is unity and the meaning of α is the same for every order.

In classical Eulerian magnification [1] and in phase-based methods [2], [3] the pipeline stops after the band-pass: the result is multiplied by the constant gain α and added back. That step is equivalent to applying D^ν with $\nu = 0$, the identity. The choice is therefore implicit and fixed, and earlier methods have no parameter taking this role. Applying D^ν with $\nu \in (0, 1]$ continuously reshapes the same band-limited signal, from the original signal up to a full temporal derivative at $\nu = 1$, progressively emphasising the upper part of the pass-band. The spectral shaping that was previously implicit becomes an explicit, continuously tunable parameter.

C. Local amplitude and phase

For each pyramid band ℓ , let b denote the band signal of dimensions $H_\ell \times W_\ell$. A computationally simple approximation of the two spatial Riesz components is obtained with first-order central differences, implemented as convolutions with the kernel $[-0.5, 0, 0.5]$ and its transpose. This gives $R_1, R_2 \in \mathbb{R}^{H_\ell \times W_\ell}$, which capture local spatial variation along the two image axes. The monogenic signal [7] then defines the local amplitude and phase as

$$A = \sqrt{b^2 + R_1^2 + R_2^2}, \quad \phi = \text{atan2}\left(\sqrt{R_1^2 + R_2^2}, b\right). \quad (5)$$

All operations are pointwise over (x, y) , so $A, \phi \in \mathbb{R}^{H_\ell \times W_\ell}$. The amplitude measures local contrast, while the phase encodes the spatial position of structure.

D. Phase-reliability adaptive gain

Since ϕ in (5) is meaningful only when A is appreciable, we define a reliability weight

$$\rho = \frac{A^2}{A^2 + \sigma^2}, \quad (6)$$

a Wiener-type shrinkage with $\rho \rightarrow 1$ near strong edges and $\rho \rightarrow 0$ in flat regions. Amplitudes well below σ therefore contribute little to the output. The threshold σ is set to the 75th percentile of the amplitude distribution of the band itself, which lets it adapt to the band content and removes the need for a hand-tuned global constant.

The effective gain is

$$\alpha_{\text{eff}} = \alpha \frac{\rho}{1 + \nu(1 - \rho)}, \quad (7)$$

which we use as a reliability-dependent shrinkage of the nominal gain. The numerator suppresses amplification where the phase is unreliable, while the denominator adds a penalty that grows with the operator order ν and disappears as $\rho \rightarrow 1$. The chosen form recovers the nominal gain α for fully reliable phase, suppresses the gain when the phase is unreliable, and reduces to the plain weighting $\alpha\rho$ when $\nu = 0$. Hence a larger ν , i.e. a stronger emphasis of high temporal frequencies, incurs a larger penalty wherever the phase is unreliable.

The amplitude used in (6) is computed from the temporal mean of each band, not from a single frame; a frame-by-frame mask would fluctuate with the noise and introduce flicker.

E. Algorithm

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1: input: frames  $\{I_t\}_{t=1}^T$ ,  $f_s$ , band  $[f_{lo}, f_{hi}]$ , gain  $\alpha$ , order  $\nu$ , levels  $L$ 
2: for  $t = 1$  to  $T$  do
3:    $P_t \leftarrow \text{LAPLACIANPYRAMID}(I_t, L)$ 
4: end for
5: for each pyramid band  $\ell$  do
6:   if  $\ell$  is the low-pass residual then
7:     leave unmodified {amplifying it would alter global scene content}
8:   else
9:      $S \leftarrow \text{STACK}(\{P_t[\ell]\}_t)$ 
10:     $\tilde{S} \leftarrow \text{FRACBANDPASS}(S, f_s, f_{lo}, f_{hi}, \nu)$ 
11:     $A \leftarrow \text{AMPLITUDE}(\text{MEAN}_t(S))$ 
12:     $\alpha_{\text{eff}} \leftarrow (7) \text{ from } A, \nu$ 
13:     $P_t[\ell] \leftarrow P_t[\ell] + \alpha_{\text{eff}} \odot \tilde{S}_t \quad \forall t$ 
14:   end if
15: end for
16: return  $\{\text{COLLAPSE}(P_t)\}_t$ 

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Here, ℓ indexes the spatial bands of the Laplacian pyramid. The low-pass residual is the coarsest, spatially smoothed component, containing large-scale scene structure and slowly varying intensity. It is left unchanged because magnifying it could alter overall scene appearance or illumination rather than the targeted temporal variations. Fig. 3 summarises the complete data flow, distinguishing spatial, temporal, frequency-domain, and fusion stages.

IV. EXPERIMENTAL SETUP

A. Why a synthetic sequence

Quantifying amplification requires knowing the true displacement, information that no real video supplies at sub-pixel precision. We therefore use a controlled sequence that also lets the two questions of interest be separated: how much genuine motion is amplified, and how much noise is injected where there is no signal to amplify.

The sequence has 96 frames of 128×128 at 30 fps. Its left half contains a horizontal sinusoid of period 16 px that oscillates sideways with amplitude 0.2 px at 2 Hz; the right half is *flat*. Additive Gaussian noise of standard deviation σ_{in} is added throughout. The displacement is well below one pixel and cannot be seen by inspection. The temporal pass-band used in the experiments is $[1.5, 2.5]$ Hz, so the 2 Hz ground-truth oscillation lies at its centre. Unless stated otherwise, all processing is performed on a single luminance channel (grayscale); the real-time colour variant processes only the Y channel of YCrCb.

B. Metrics

Amplification factor: motion amplification is quantified by comparing the displacement estimated from the processed video with the known ground-truth displacement. For each frame, displacement is estimated with aggregate 1-D optical flow relative to the mean frame. Under brightness constancy

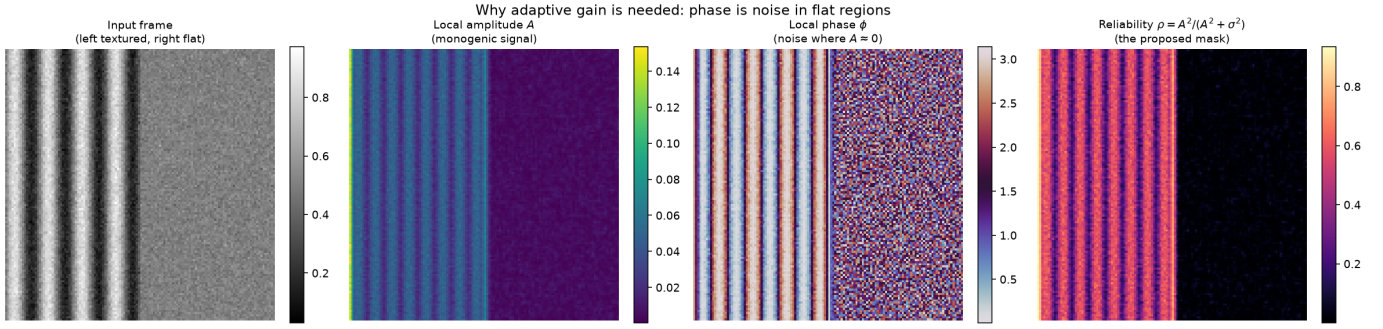


Fig. 2. Motivation for the adaptive gain. The input (left) is textured on the left half and flat on the right. Local phase (third panel) is noise in the flat region, where amplitude vanishes; the reliability mask (6) (right) identifies this without supervision.

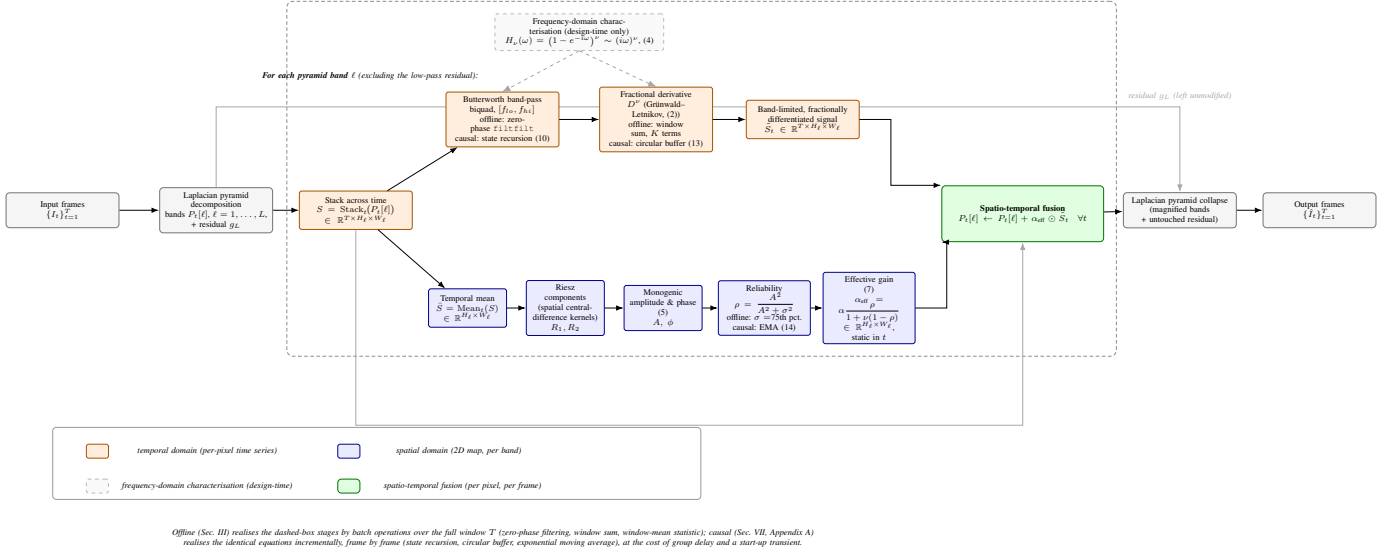


Fig. 3. End-to-end data flow of FrAM for one pyramid band ℓ . The loop over bands runs independently for every scale, and the low-pass residual bypasses it completely. After the temporal stack S the pipeline splits into a temporal/frequency branch that produces the fractionally differentiated signal $\tilde{S}_t$, and a spatial branch computed once from the band's temporal mean that produces the static gain map α_{eff} ; the two are fused with the original unfiltered band. Stages with both offline and causal realisations are annotated with both modes (Sec. VII, Appendix A).

and for small displacements, $I_t + dI_x \approx 0$; the least-squares solution over all pixels is

$$d = -\frac{\sum I_t I_x}{\sum I_x^2}.$$

After removing the temporal mean from each series, a through-origin least-squares fit

$$d_{\text{est}} = a d_{\text{GT}}, \quad a = \frac{\sum d_{\text{GT}} d_{\text{est}}}{\sum d_{\text{GT}}^2},$$

defines the **amplification factor** as the slope a . Thus $a = 1$ means faithful motion estimation, $a > 1$ means amplification, and $a < 1$ means attenuation. Applied to the unprocessed input, the estimator gives $a = 1.01$, indicating a negligible intrinsic bias.

Regional temporal noise: to separate amplified sensor noise from amplified motion, temporal intensity fluctuation is measured independently in the flat and textured halves. Let $\Delta I_t(\mathbf{x}) = I_{t+1}(\mathbf{x}) - I_t(\mathbf{x})$ be the inter-frame difference.

Over a spatial region $\mathcal{R}$ the noise score is the pooled standard deviation

$$\sigma_{\mathcal{R}} = \text{std}\{\Delta I_t(\mathbf{x}) : t = 1, \dots, T-1, \mathbf{x} \in \mathcal{R}\}.$$

The flat region uses $\mathcal{R}_{\text{flat}} = \{x \geq 70\}$ and the textured region $\mathcal{R}_{\text{tex}} = \{x < 58\}$, leaving a margin around the texture–flat boundary at $x = 64$. Consecutive-frame differencing high-pass filters the time series, so $\sigma_{\mathcal{R}}$ emphasises temporal mottling rather than static spatial texture; in the flat half it is therefore a direct readout of injected noise.

Band SNR: spectral fidelity of the recovered motion is measured on the same displacement series $d_{\text{est}}(t)$ used for the amplification factor. After mean removal we form the one-sided periodogram $P(f) = |\hat{d}_{\text{est}}(f)|^2$ via the real DFT and

TABLE I

COMPARISON WITH α CALIBRATED TOWARD EQUAL EFFECTIVE AMPLIFICATION ($\text{EVM } 18.44\times$), $\sigma_{\text{IN}} = 0.04$. FRAM FALLS SLIGHTLY SHORT OF THE EVM FACTOR BECAUSE THE RELIABILITY MASK WITHHOLDS GAIN IN THE FLAT HALF; LOWER NOISE IS BETTER. BOLD MARKS THE BETTER VALUE OF EACH METRIC COLUMN AMONG THE PROCESSED METHODS (NOISE, SNR); THE AMPLIFICATION SERVES AS THE CALIBRATION TARGET, AND THE UNPROCESSED INPUT IS LISTED FOR REFERENCE ONLY.

Method	α	Amplif.	Noise (flat)	Noise (textured)	SNR (dB)
Input (unprocessed)	–	$1.01\times$	0.0565	0.0569	8.86
EVM (Butterworth)	20	$18.44\times$	0.1137	0.1589	11.99
FrAM ($\nu = 0.0$)	42	$17.03\times$	0.0574	0.1525	11.89
FrAM ($\nu = 0.3$)	64	$16.09\times$	0.0574	0.1589	11.91
FrAM ($\nu = 0.5$)	90	$14.83\times$	0.0575	0.1733	11.74
FrAM ($\nu = 0.7$)	131	$12.77\times$	0.0576	0.1987	11.41

define

$$\text{SNR}_{\text{band}} = 10 \log_{10} \frac{P_{\text{in}}}{P_{\text{out}}},$$

$$P_{\text{in}} = \sum_{|f-f_s| < \Delta f} P(f),$$

$$P_{\text{out}} = \sum_{\substack{f > 0 \\ |f-f_s| \geq \Delta f}} P(f),$$

with signal frequency $f_s = 2\text{ Hz}$ and half-width $\Delta f = 0.5\text{ Hz}$, matching the synthetic oscillation. Spatial intensity variance is deliberately *not* used: under a near-rigid translation it is almost constant in time and carries no oscillatory signature. A higher SNR_{band} indicates that the magnified motion concentrates energy at f_s rather than spreading it into broadband temporal noise.

C. Protocol

Methods are compared with α **calibrated toward equal effective amplification**. Because the pipeline is linear in α under a spatially uniform gain, each configuration's α is scaled by the ratio of the baseline amplification to a probe run. With the reliability mask the realised factor can remain slightly below the baseline, because flat pixels receive little gain. This is the only fair comparison: any method would look less noisy if it simply amplified less.

V. RESULTS

A. Main result

Table I reports the main comparison. With α calibrated toward the Eulerian factor, FrAM holds flat-region temporal noise at the input level (0.0574, versus 0.0565 for the input), while the Eulerian baseline doubles it to 0.1137; the output therefore remains clean in regions where the baseline injects spurious motion. The residual gap in measured amplification ($17.0\times$ at $\nu = 0$ versus $18.4\times$) is the expected cost of withholding gain where there is no structure to amplify.

The textured region behaves differently depending on ν . At $\nu \leq 0.3$ FrAM matches the baseline (0.1589 for both at $\nu = 0.3$); raising ν increases textured-region noise, as (4)

TABLE II

ADAPTIVE-GAIN ABLATION, $\alpha = 20$ FIXED. WITHIN EACH ORDER, BOLD MARKS THE GAIN WITH THE LOWER NOISE IN EACH NOISE COLUMN. AT FIXED α THE ADAPTIVE GAIN ALWAYS LOWERS NOISE AND ALWAYS LOWERS AMPLIFICATION; MATCHING AMPLIFICATION REQUIRES RECALIBRATING α (TABLE I).

Order	Gain	Amplif.	Noise (flat)	Noise (textured)
$\nu = 0.0$	constant	$18.44\times$	0.1137	0.1589
$\nu = 0.0$	adaptive	$8.75\times$	0.0569	0.0925
$\nu = 0.5$	constant	$10.05\times$	0.0935	0.1188
$\nu = 0.5$	adaptive	$4.11\times$	0.0567	0.0723
$\nu = 0.7$	constant	$6.70\times$	0.0876	0.1072
$\nu = 0.7$	adaptive	$2.81\times$	0.0566	0.0679

TABLE III

ROBUSTNESS TO INPUT NOISE, AT MATCHED AMPLIFICATION, $\nu = 0.3$. BOLD MARKS THE LOWER FLAT-REGION NOISE OF THE TWO METHODS ON EACH ROW; THE AMPLIFICATION COLUMNS RECORD THE CALIBRATION TARGET, NOT A QUANTITY TO MAXIMISE.

σ_{in}	EVM		FrAM	
	Amplif.	Noise (flat)	Amplif.	Noise (flat)
0.01	$19.15\times$	0.0284	$16.81\times$	0.0142
0.02	$18.98\times$	0.0569	$16.64\times$	0.0284
0.04	$18.44\times$	0.1137	$16.09\times$	0.0574
0.08	$16.64\times$	0.2275	$14.36\times$	0.1207

predicts, because higher order emphasises the high frequencies where noise dominates. Together with the loss of amplification at high ν , this points to $\nu \in [0, 0.3]$ as the useful operating range.

B. Ablation

Table II separates the contribution of the two components. With a constant gain, increasing ν does reduce noise, but at a severe cost in amplification ($18.44\times \rightarrow 6.70\times$). The adaptive gain, by contrast, brings flat-region noise down to the input level at every order. The noise reduction is therefore attributable to the adaptive gain, not to the fractional order, whose role is to control the amplification–noise trade-off in textured regions.

C. Robustness

Table III shows that the behaviour is stable across an eightfold range of input noise: FrAM roughly halves flat-region noise at every level while retaining approximately 87% of the baseline amplification.

Fig. 4 presents the same result visually. The spatio-temporal slices show the flat region of the baseline output populated with spurious temporal texture, absent from FrAM. The displacement traces confirm that neither method introduces phase lag, as expected from zero-phase filtering.

VI. VALIDATION ON REAL VIDEO

The synthetic experiment quantifies behaviour under controlled conditions but does not establish that the method is useful on real material. We therefore evaluate on the four sequences distributed with the reference implementation of [1]:

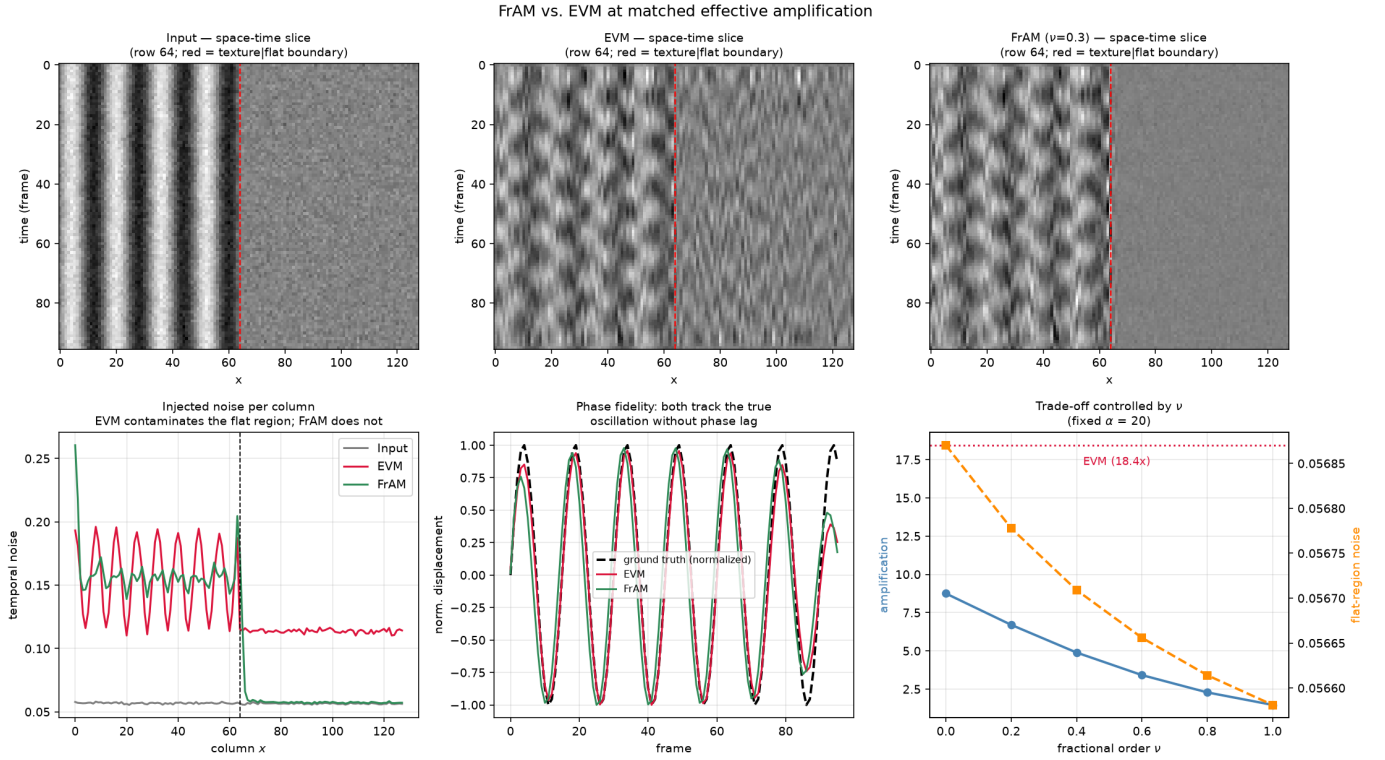


Fig. 4. Comparison at equal effective amplification. Top: spatio-temporal slices; right of the dashed line the region is flat, and the baseline has filled it with spurious texture that FrAM leaves untouched. Bottom left: injected noise per column, dropping sharply at the boundary for FrAM. Bottom centre: both methods track the true oscillation without phase lag. Bottom right: the trade-off controlled by ν .

face and face2 (facial pulse), baby (infant respiration) and subway (structural vibration). Each clip was processed as a single grayscale channel at half resolution (or $0.35\times$ for all but face), using the first 180 frames.

A. Evaluation without ground truth

Real video provides no ground-truth sub-pixel displacement, so the amplification factor of Section IV cannot be computed. We substitute three criteria that remain objective:

(1) **Physiological selectivity.** In the facial sequences the pulse lies in a known band (0.7–2.0 Hz, the band used for face/face2; baby uses 0.5–2.0 Hz and subway 1.0–4.0 Hz). We report the band energy ratio (BER), the fraction of temporal energy falling inside the target band, and the dominant frequency in beats per minute.

(2) **Noise in signal-free regions.** Each scene is segmented *without supervision* into an active and a static region by thresholding the per-pixel temporal standard deviation at the 80th and 30th percentiles. The masks are computed on the *original* video, so they are identical for every method compared. Any temporal noise injected into the static region is pure artefact.

(3) **Spatial selectivity.** The ratio of enhancement in the active region to enhancement in the static region. A value of unity indicates indiscriminate amplification.

Methods are compared at **equal enhancement in the active region**: FrAM’s α is scaled until its enhancement ratio in

TABLE IV
REAL VIDEO, AT EQUAL ENHANCEMENT IN THE ACTIVE REGION. WITHIN EACH VIDEO, BOLD MARKS THE BETTER OF EVM AND FRAM ON EACH METRIC (HIGHER BER, LOWER BACKGROUND NOISE, HIGHER SPATIAL SELECTIVITY); EVM RETAINS SLIGHTLY HIGHER BER ON THREE OF THE FOUR CLIPS. THE UNPROCESSED INPUT IS SHOWN ONCE AS A REFERENCE, AND THE PULSE RATE IS A CONSISTENCY CHECK RATHER THAN A QUANTITY TO MAXIMISE.

Video	Method	BER	Noise (bg)	Spatial select.	Rate (bpm)
face	input	0.136	0.00233	1.00	50
face	EVM	0.825	0.00909	1.40	50
face	FrAM	0.844	0.00302	4.53	50
face2	EVM	0.332	0.00702	1.14	50
face2	FrAM	0.244	0.00662	1.31	70
baby	EVM	0.395	0.00468	1.65	–
baby	FrAM	0.362	0.00246	3.49	–
subway	EVM	0.742	0.02026	1.16	–
subway	FrAM	0.598	0.00780	3.19	–

the active region matches the EVM one, as in the synthetic protocol.

B. Results

Table IV shows that FrAM reduces background noise on all four sequences, by a margin that varies considerably. On face the reduction is $3.0\times$ (0.00302 versus 0.00909) while in-band energy is preserved (BER = 0.844 versus 0.825) and

TABLE V

ACTIVITY CONTRAST OF THE INPUT AGAINST THE OBSERVED ADVANTAGE. BOLD MARKS THE LOWER BACKGROUND NOISE OF THE TWO METHODS ON EACH ROW; THE GAIN COLUMN IS THE RATIO OF THE TWO NOISE COLUMNS. THE ORDERING IS NOT MONOTONIC: CONTRAST DOES NOT PREDICT THE MARGIN.

Video	Activity contrast	Noise (bg) EVM	Noise (bg) FrAM	Gain
subway	6.04	0.02026	0.00780	2.60×
face2	3.77	0.00702	0.00662	1.06×
face	1.98	0.00909	0.00302	3.01×
baby	1.07	0.00468	0.00246	1.90×

the same pulse rate is recovered; spatial selectivity rises from 1.40 to 4.53. On *subway* background noise falls by 2.6× and selectivity from 1.16 to 3.19. On *baby* the reduction is 1.9×. On *face2* the advantage is marginal (1.06×).

The consistent pattern across sequences is spatial selectivity: FrAM raises it in every case, most sharply on *face* and *subway*. This is the behaviour the reliability weighting is designed to produce, and it is the quantity least sensitive to how the enhancement is calibrated.

Define the *activity contrast* of a sequence as the ratio of temporal noise in the active region to that in the static region of the *original* video. Table V shows that this quantity does *not* order the outcomes: the largest gain occurs at a contrast of 1.98 and the smallest at 3.77, and the linear correlation over the four sequences is negligible ($r = 0.03$). Four sequences cannot support a predictor of the margin, and we offer none.

The advantage is smallest on *face2*, whose input carries the least in-band energy of the four (BER = 0.022, roughly six times below the other three). Where the band-pass isolates little genuine signal, both methods amplify predominantly noise and the reliability mask has correspondingly little to distinguish. We regard this as a plausible mechanism, not a validated criterion: settling an applicability rule would require far more than four clips.

VII. REAL-TIME FORMULATION

A. Diagnosis of the cost

The pipeline of Section III, applied to a sliding window, costs over 2 s per output frame at 640×480 with $T = 32$ (2430 ms in Table VI). Profiling attributes the time almost entirely to the temporal filter:

Stage	Cost
Laplacian pyramids ($T = 32$ frames)	~90 ms
Temporal filter	~1.95 s
Pyramid collapse ($T = 32$ frames)	~50 ms

Three sources of inefficiency compound here. First, the window is reprocessed in full at every step and all but the newest frame is discarded. Second, *filtfilt* traverses the sequence twice to obtain zero phase. Third, the convolution (2) is evaluated as K shifts of the whole $T \times H \times W$ volume.

TABLE VI

PER-FRAME COST: OFFLINE VERSUS CAUSAL STREAMING. STREAMING IS CHEAPER AT EVERY RESOLUTION TESTED; BOLD HIGHLIGHTS THE 640×480 ROW, THE REFERENCE CONFIGURATION QUOTED ELSEWHERE IN THIS PAPER.

Resolution	Offline	Streaming	Speed-up	fps
320×240	382 ms	6.3 ms	60×	158
640×480	2430 ms	14.6 ms	166×	69
1280×720	7772 ms	62.6 ms	124×	16
1920×1080	17937 ms	108.8 ms	165×	9

B. Causal reformulation

None of the three is inherent to fractional calculus; all arise from processing a window instead of a stream. The same operator admits a causal, incremental form built on three substitutions, detailed in Appendix A:

- 1) The band-pass of Section III is evaluated as a second-order state recursion (10), keeping two past inputs and outputs per pixel. This is the identical filter, not an approximation: against *lfilter* the maximum discrepancy is 6×10^{-16} , i.e. machine precision. Cost per frame drops from $O(T)$ to $O(1)$.
- 2) The truncated Grünwald–Letnikov sum (2) is evaluated over a circular buffer of $K + 1$ band images, as one weighted combination rather than K shifts of a $T \times H \times W$ volume.
- 3) The reliability statistic of (6) is updated by an exponential moving average instead of being recomputed over the window mean.

Causality carries two costs: the filter has non-zero phase, so it introduces group delay where the offline mode has none, and roughly 40 frames of transient pass before the output is valid.

C. Results

Table VI reports the outcome: a 166× speed-up at 640×480 (69 fps), crossing the 30 fps threshold at QVGA and VGA. A colour variant operating on the luminance channel of YCrCb, leaving chrominance untouched, costs 18.5 ms (54 fps at 640×480): the magnification work is confined to one channel, so its cost remains close to grayscale. The timings in Table VI come from a single run and are representative rather than exact: repeated executions on the same machine give values that can differ by tens of percent. The dependable outcome is the order-of-magnitude gap between the two modes, not any individual ratio.

Table VII confirms that the advantage survives the reformulation, attenuated: flat-region noise falls from 0.1605 to 0.1091, a factor of 1.47× against the 2× of the offline mode. The cause is that the causal mask, lacking the average over the full window, is noisier than its offline counterpart.

Table VIII shows how much fractional memory is needed. Truncating to $K = 8$ yields the same amplification as $K = 31$ at roughly 60% of its cost in this run. The long-memory property of the operator matters for its spectral character (the

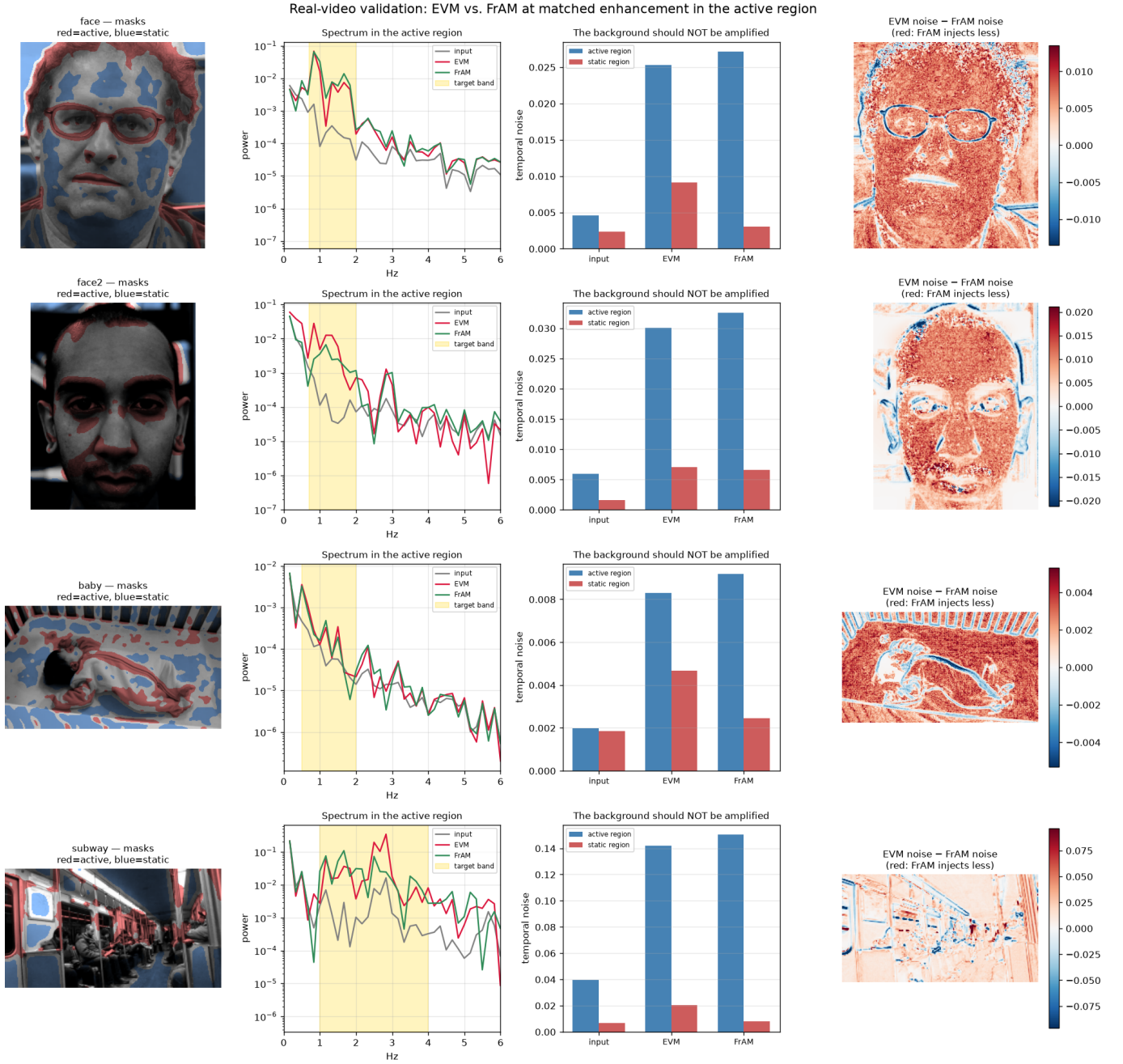


Fig. 5. Real-video validation. Columns: unsupervised region masks; temporal spectrum of the active region with the target band shaded; temporal noise per region; and the difference map of injected noise (red where FrAM injects less). Red dominates on all four sequences, most clearly for face and subway.

$|\omega|^\nu$ response of (4)), but the far tail contributes little to the measured result, so the memory noted in Sec. III-A turns out to be largely unnecessary in practice.

VIII. LIMITATIONS

Attribution. As Table II establishes, the noise reduction follows from the adaptive gain; the fractional operator contributes the continuous control parameter, not the improvement. A reader interested only in the noise result could adopt (7) with $\nu = 0$.

Variable margin, unexplained. The background-noise reduction ranges from $1.06\times$ to $3.01\times$ across four sequences, and Table V shows that the activity contrast of the input does not predict where in that range a given sequence falls. We can offer a mechanism for the weakest case but not a criterion, and four clips are too few to establish one. Which scenes benefit most therefore remains open.

Ambiguity on face2. The gain is marginal and the dominant frequency shifts from 50 to 70 bpm. Both lie within the physiological range, but the discrepancy should be resolved

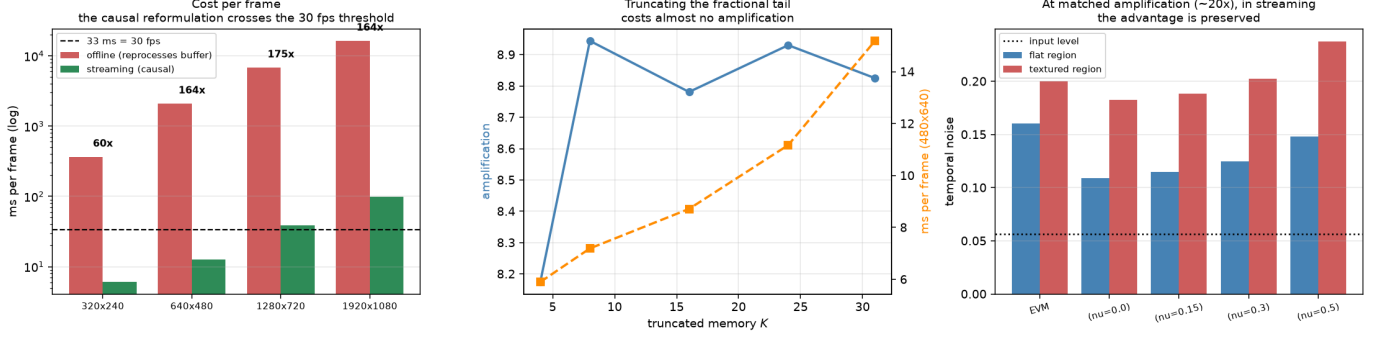


Fig. 6. Causal reformulation. Left: per-frame cost, log scale, with the 30 fps threshold marked. Centre: truncating the fractional memory costs little amplification while cost grows with K . Right: at equal amplification in streaming, the advantage over the baseline is retained.

TABLE VII

STREAMING, AT EQUAL EFFECTIVE AMPLIFICATION. BOLD MARKS THE LOWEST NOISE OF EACH COLUMN AMONG THE PROCESSED METHODS; THE UNPROCESSED INPUT IS LISTED FOR REFERENCE ONLY.

Method	α	Amplif.	Noise (flat)	Noise (text.)
Input	–	$1.01\times$	0.0565	0.0569
EVM (streaming)	20	$20.85\times$	0.1605	0.2002
FrAM ($\nu = 0.0$)	33	$20.08\times$	0.1091	0.1827
FrAM ($\nu = 0.15$)	39	$19.78\times$	0.1149	0.1887
FrAM ($\nu = 0.3$)	47	$19.36\times$	0.1250	0.2026
FrAM ($\nu = 0.5$)	64	$18.47\times$	0.1480	0.2375

TABLE VIII

TRUNCATING THE FRACTIONAL MEMORY (640×480). AMPLIFICATION AND FLAT-REGION NOISE VARY LITTLE FOR $K \geq 8$, WHILE COST GROWS WITH K ; BOLD HIGHLIGHTS THE TRUNCATION $K = 8$ USED BY THE INTERACTIVE VIEWER.

K	Amplif.	Noise (flat)	ms/frame
4	$8.17\times$	0.0779	7.8
8	$8.94\times$	0.0783	11.2
16	$8.78\times$	0.0781	12.1
24	$8.93\times$	0.0782	15.0
31	$8.82\times$	0.0781	17.9

against a reference measurement such as contact photoplethysmography.

Two modes with different properties. The offline mode uses a zero-phase filter and yields the $2\times$ noise reduction; the causal mode yields $1.47\times$. The figures reported for FrAM in this paper are those of the offline mode and should not be quoted as streaming numbers.

First-order Riesz approximation. The kernels $[-0.5, 0, 0.5]$ are a first-order approximation; a full steerable-filter implementation would estimate phase more accurately.

Global order. A single ν is used for all bands. Since noise characteristics differ across scales, a per-band order is a natural extension.

IX. CONCLUSION

We presented FrAM, a motion-magnification method that combines a fractional-order Grünwald–Letnikov temporal fil-

ter with a gain weighted by the reliability of the local phase. On a synthetic sequence with known sub-pixel ground truth, the method approaches the amplification of classical Eulerian magnification while holding temporal noise in textureless regions at the input level, where the Eulerian baseline doubles it, and this holds across an eightfold range of input noise. On real sequences the effect is confirmed: background noise falls by $3.0\times$ on the primary facial-pulse clip and $2.6\times$ on the structural-vibration clip, with in-band energy and pulse rate preserved, and spatial selectivity improves on every sequence. The noise reduction, however, ranges from $1.06\times$ to $3.01\times$ across the four clips by a mechanism we could not identify. An ablation attributes the improvement to the adaptive gain and characterises the fractional order as a control parameter for the amplification–noise trade-off. The reliability weighting is independent of the rest of the pipeline and could be applied to existing phase-based magnification methods directly.

We further showed that the computational cost of the formulation was an artefact of implementation rather than of the fractional operator. Recasting the band-pass as a state recursion (10) and the Grünwald–Letnikov sum over a circular buffer (13) reduces the per-frame cost by $166\times$, reaching 69 fps in grayscale and 54 fps in colour at 640×480 , and retains a $1.47\times$ noise reduction over the baseline. Truncating the fractional memory to $K = 8$ costs no measurable amplification, which suggests the long-memory property matters for the spectral character of the operator rather than for the magnitude of the effect.

REPRODUCIBILITY

Source code, the interactive real-time viewer and result videos are available at <https://github.com/agarnung/FrAM-Motion-Magnification>.

The implementation depends only on NumPy, SciPy, OpenCV and Matplotlib; no GPU, deep-learning framework or specialised pyramid library is required. Every table and figure in this paper is produced by one of three scripts, with fixed parameters and fixed random seeds where randomness is involved: `experiments.py` for the synthetic evaluation of Tables I–III, `real_video.py` for the real-video validation

of Tables IV–V, and `experiments_rt.py` for the streaming results of Tables VI–VIII. The two magnification engines are separate modules: `fracmag.py` implements the offline, zero-phase formulation of Section III, and `fracmag_rt.py` the causal streaming formulation of Section VII and Appendix A. The real sequences used in Section VI are those distributed with the reference implementation of [1]. All experiments process a single luminance or grayscale channel; the real-time colour stream converts the input to YCrCb and magnifies only the Y channel.

APPENDIX A

IMPLEMENTATION OF THE CAUSAL FORMULATION

This appendix collects the implementation details behind Section VII. It is separated from the body because it concerns realisation rather than method: the operator being computed is the one defined in Section III.

A. Band-pass as a state recursion

The band-pass filter of Section III is a biquad. This follows from how `scipy.signal.butter` realises a bandpass design: it builds an order- N lowpass *prototype* and reaches the bandpass response through the standard lowpass-to-bandpass substitution

$$s \mapsto \frac{s^2 + \omega_0^2}{sB}, \quad (8)$$

where $\omega_0 = \sqrt{\omega_{lo}\omega_{hi}}$ is the centre frequency and $B = \omega_{hi} - \omega_{lo}$ the bandwidth. Substituting (8) into each simple pole of the prototype introduces an s^2 term in the denominator, so a single pole becomes a conjugate *pair* of poles: the order doubles from N to $2N$. For $N = 1$, with prototype $H_{lp}(s) = 1/(s + 1)$, this gives

$$H_{bp}(s) = H_{lp}\left(\frac{s^2 + \omega_0^2}{sB}\right) = \frac{sB}{s^2 + Bs + \omega_0^2}, \quad (9)$$

an analog transfer function with two poles and one zero (at $s = 0$). The bilinear transform that discretises (9) maps the point at infinity to $z = -1$, contributing a second zero; the resulting discrete transfer function therefore has two poles *and* two zeros, i.e. three coefficients b_0, b_1, b_2 and three coefficients a_0, a_1, a_2 ($a_0 = 1$ after normalisation); this is a biquad. This is why `butter(1, [lo, hi], 'bandpass')`, despite the “order 1” argument, returns coefficient arrays of length three rather than two: the argument sets the order of the lowpass prototype, not of the realised bandpass filter. Writing b_i, a_i for these coefficients, the output at frame n is

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2], \quad (10)$$

evaluated elementwise over the band image, with the four state arrays $x[n-1], x[n-2], y[n-1], y[n-2]$ retained between frames. Since (10) is the direct-form realisation of the same transfer function implemented by `scipy.signal.lfilter`, the state recursion reproduces the output of a single forward pass of that routine up to floating-point error; we verified a maximum deviation of 6×10^{-16} over a 60-frame sequence. The offline formulation

uses `filtfilt`, which applies the same filter forward and backward with carefully chosen initial conditions to achieve zero phase; the causal streaming mode cannot replicate those initial conditions and therefore introduces group delay.

B. Fractional sum over a circular buffer

This step does not change the Grünwald–Letnikov derivative itself, only how the $K + 1$ frames it sums over are stored and addressed. Let $\mathbf{y}_n$ denote the band-pass output at frame n and w_k the weights of (3). As in (2), the truncated sum needed at frame n is

$$D^\nu \mathbf{y}_n \approx \sum_{k=0}^K w_k \mathbf{y}_{n-k}, \quad (11)$$

i.e. it requires the current output together with its K most recent predecessors, $\mathbf{y}_n, \mathbf{y}_{n-1}, \dots, \mathbf{y}_{n-K}$, each multiplied by its corresponding weight $w_0, w_1, \dots, w_K$. A naïve sliding-window implementation keeps these $K + 1$ frames as an explicit array and shifts every entry down by one position after each frame, so that $\mathbf{y}_{n-k}$ is always found at the same offset; this touches the whole $T \times H \times W$ volume at every step, which is the cost the causal formulation is meant to remove.

The circular buffer removes the shift by relaxing where each frame is stored, and compensating with where it is read from. The $K + 1$ frames are kept in a fixed array of $K + 1$ slots, $B[0], \dots, B[K]$, addressed through a pointer p that is updated in two steps whenever a new frame arrives: $\mathbf{y}_n$ is written into the slot p currently points to, and only *then* is p advanced, wrapping back to 0 once it would pass the last slot: $p \leftarrow (p + 1) \bmod (K + 1)$, so that p takes exactly the slot indices $0, \dots, K$.

The same wrap-around is needed when reading a past frame, and gives the index map below. Because writing happens *before* advancing, the slot just written, holding $\mathbf{y}_n$, sits at $(p - 1)$ rather than at p ; the frame written the step before, $\mathbf{y}_{n-1}$, sits at $(p - 2)$, and the k -th most recent frame $\mathbf{y}_{n-k}$ at $(p - 1 - k)$. Taken as an ordinary integer, $p - 1 - k$ can run negative (e.g. $p = 0, k = 2$ gives -3) or wrap past 0 repeatedly as k grows, so it is folded back into the valid range $\{0, \dots, K\}$ with the same $\bmod(K + 1)$:

$$\mathbf{y}_{n-k} = B[(p - 1 - k) \bmod (K + 1)], \quad (12)$$

where p is understood *after* the write-then-advance step for frame n . Concretely, with $K = 3$ (four slots, $B[0..3]$): if the frame $\mathbf{y}_n$ just written ends up leaving $p = 0$ (it was written into slot 3, the last one, and p then wrapped past it), the map (12) places $\mathbf{y}_n$ at $B[3]$ ($k=0$: $(0-1-0) \bmod 4 = -1 \bmod 4 = 3$), $\mathbf{y}_{n-1}$ at $B[2]$ ($k=1$: $-2 \bmod 4 = 2$), $\mathbf{y}_{n-2}$ at $B[1]$ ($k=2$: $-3 \bmod 4 = 1$), and $\mathbf{y}_{n-3}$ at $B[0]$ ($k=3$: $-4 \bmod 4 = 0$): the four stored frames are addressed backwards from p , with no data ever copied between slots. Substituting (12) into (11) gives

$$D^\nu \mathbf{y}_n \approx \sum_{k=0}^K w_k B[(p - 1 - k) \bmod (K + 1)], \quad (13)$$

i.e. one weighted sum of $K + 1$ stored images, with the index arithmetic alone guaranteeing that the newest entry is always paired with w_0 , the second-newest with w_1 , and so on, matching the ordering assumed in (11). Each new frame therefore costs one write into the slot p pointed to before it is advanced, followed by the weighted sum (13), rather than K shifts of a $T \times H \times W$ volume. On the test machine the incremental cost of this weighted sum is a few milliseconds per frame at 640×480 with $K = 31$, far below the cost of reprocessing the whole temporal window.

C. Incremental reliability mask

The offline mask of (6) takes its amplitude from the temporal mean of each band, which is unavailable causally. It is replaced by an exponential moving average,

$$\bar{A}_n = \tau \bar{A}_{n-1} + (1 - \tau) A_n, \quad \tau = 0.9, \quad (14)$$

and the threshold σ is refreshed from a percentile of $\bar{A}_n$ every eighth frame and then itself smoothed; recomputing a percentile at every frame would cost time and make the mask fluctuate visibly.

Because $\bar{A}_n$ is built from instantaneous bands rather than a temporal mean, it carries more noise than its offline counterpart. To compensate, the causal mask computes the quadrature pair with Sobel kernels, which factor as $[1, 2, 1]^\top * [-1, 0, 1]$: up to a constant factor, the differencing part coincides with the first-order Riesz kernels $[-0.5, 0, 0.5]$ of (5), while the transverse smoothing attenuates noise before the amplitude is formed. This regularisation is a deliberate divergence from the offline definition: the causal EMA of an instantaneous band needs the extra smoothing to keep the mask stable. The cost of computing the mask is sub-millisecond per frame at 640×480 , comparable to that of the offline Riesz-kernel variant.

D. Numerical precision and colour

All state and arithmetic are single precision. Measured on the recursion (10) at 640×480 , `float32` costs 0.31 ms per frame against 1.78 ms for `float64`, with no perceptible difference in the output; the gain is bandwidth, not arithmetic.

The colour variant converts to YCrCb and applies the pipeline to the luminance channel alone, leaving chrominance untouched apart from an optional saturation factor. Since the magnification acts on geometry rather than colour, this preserves the effect while keeping the cost close to the grayscale case (18.5 ms for the colour stream against 14.6 ms for grayscale at 640×480 ; run-to-run timing noise alone spans this difference).

E. Parameter invalidation

Amplification α , the percentile, the adaptive flag and the method switch enter only as scalars in the final combination, so they can change between frames with no state rebuild. The band limits, the order ν , the number of pyramid levels, K , the colour mode and the frame size all determine filter coefficients or buffer shapes, and changing any of them discards and rebuilds the state. The viewer keys these on a tuple so that interactive slider movements of the first group are free.

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